STM32F446RET6 board: parts list

Every part, by schematic sheet, with the LCSC number to search in EasyEDA

Four sheets, one project. A net label with the same name is the same wire on every sheet.

1 Open the library
Press Shift + F, or click Library in the left panel.

2 Paste the LCSC number
Type it in the search box, for example **C14663**, and press Enter.

3 Check the match
Pick the LCSC / JLCPCB result whose MFR part matches this list.

4 Place it
Click Place, then click on the sheet. Edit › Annotate renumbers C1, R1 and the rest.

Search the C-number, not the value. "100nF" finds thousands of parts. "C14663" finds one, with its footprint and 3D model linked, and it's the number JLCPCB assembles from.

Basic or Extended? Each different Extended part adds a loading fee if JLCPCB assembles the board. **Check**: confirm it in the search result. Hand-soldering? Ignore it.

Ref	Part and value	Package	LCSC # · MFR part	JLC class	Qty	Connects
Sheet_1 MCU core	U1 and everything that sits right next to its pins					
☐ **U1**	**Microcontroller** 180 MHz · 512 KB flash	LQFP-64	**C69336** STM32F446RET6	Extended	1	Every pin: see the build sheet
☐ **C1–C4**	**Decoupling cap** 100 nF · 50 V · X7R	0603	**C14663** CC0603KRX7R9BB104	Basic	4	One at each VDD pin (19, 32, 48, 64) to GND
☐ **C5**	**VBAT cap** 100 nF	0603	**C14663** CC0603KRX7R9BB104	Basic	1	VBAT (pin 1, on +3V3) to GND
☐ **C6**	**Bulk cap** 4.7 µF · 16 V · X5R	0603	**C19666** CL10A475KO8NNNC	Basic	1	+3V3 to GND
☐ **FB1**	**Ferrite bead** 600 Ω at 100 MHz · 200 mA	0603	**C1002** GZ1608D601TF	Basic	1	+3V3 to VDDA
☐ **C7**	**VDDA cap** 1 µF · 50 V · X5R	0603	**C15849** CL10A105KB8NNNC	Basic	1	VDDA (pin 13) to GND
☐ **C8**	**VDDA cap** 100 nF	0603	**C14663** CC0603KRX7R9BB104	Basic	1	VDDA to GND
☐ **C9**	**VCAP cap** 4.7 µF ceramic (ESR under 1 Ω)	0603	**C19666** CL10A475KO8NNNC	Basic	1	VCAP (pin 30) to GND. **Never to 3.3 V.**
☐ **C10**	**Reset filter cap** 100 nF	0603	**C14663** CC0603KRX7R9BB104	Basic	1	NRST (pin 7) to GND
☐ **R1**	**BOOT0 pull-down** 10 kΩ · 1%	0603	**C25804** 0603WAF1002T5E	Basic	1	BOOT0 (pin 60) to GND
☐ **SW1**	**Reset button** Tactile, 5.1 × 5.1 mm	SMD-4P	**C318884** TS-1187A-B-A-B	Check	1	NRST to GND
Sheet_2 Power and USB	USB-C in, 5 V to 3.3 V, USB data to PA11 and PA12					
☐ **J1**	**USB-C receptacle** USB 2.0 · 16 contacts	SMD, right angle	**C165948** TYPE-C-31-M-12	Extended	1	D+ ×2 to USB_DP, D− ×2 to USB_DM, VBUS to VBUS, shell to GND
☐ **R2, R3**	**CC pull-downs** 5.1 kΩ · 1%	0603	**C23186** 0603WAF5101T5E	Basic	2	One each: CC1 to GND, CC2 to GND
☐ **U3**	**USB ESD protection** 2 data lines + VBUS clamp	SOT-23-6	**C7519** USBLC6-2SC6	Extended	1	1, 6 USB_DM · 3, 4 USB_DP · 5 VBUS · 2 GND
☐ **U2**	**3.3 V regulator (LDO)** 3.3 V · 500 mA	SOT-23-5	**C82942** ME6211C33M5G-N	Extended	1	1 VIN, 3 CE VBUS · 2 GND · 5 VOUT +3V3 · 4 open
☐ **C15**	**Regulator input cap** 10 µF · 10 V · X5R	0603	**C19702** CL10A106KP8NNNC	Basic	1	VBUS to GND, at U2 pin 1
☐ **C16**	**Regulator output cap** 10 µF · 10 V · X5R	0603	**C19702** CL10A106KP8NNNC	Basic	1	+3V3 to GND, at U2 pin 5
☐ **R5**	**Power LED resistor** 1 kΩ · 1%	0603	**C21190** 0603WAF1001T5E	Basic	1	+3V3 to D2 anode
☐ **D2**	**Power LED (optional)** Red	0603	**C2286** KT-0603R	Basic	1	Cathode to GND. Lit = 3.3 V is up

Parts list, continued

Sheets 3 and 4, then one line per part to order

Checked on jlcpcb.com part pages, Oct 31, 2026. Stock and class change, so confirm them when you place each part.

Ref	Part and value	Package	LCSC # · MFR part	JLC class	Qty	Connects
Sheet_3 Clocks and debug	The two crystals and the SWD header					
☐ **Y1**	**8 MHz crystal (HSE)** C_L 20 pF · ±10 ppm · ESR 80 Ω	SMD5032	**C115962** X50328MSB2GI	Basic	1	Between OSC_IN (pin 5) and OSC_OUT (pin 6)
☐ **C11, C12**	**HSE load caps** 27 pF · 50 V · C0G	0603	**C1656** CL10C270JB8NNNC	Extended	2	OSC_IN to GND, OSC_OUT to GND
☐ **Y2**	**32.768 kHz crystal (LSE), optional** C_L 6 pF · ±20 ppm · ESR 50 kΩ	SMD, 4 pads	**C1985642** ABS25-32.768KHZ-6-T	Check	1	Between OSC32_IN (pin 3) and OSC32_OUT (pin 4)
☐ **C13, C14**	**LSE load caps, optional** 4.3 pF · 50 V · C0G	0603	**C59526** 0603CG4R3C500NT	Check	2	OSC32_IN to GND, OSC32_OUT to GND
☐ **J2**	**SWD header** 1 × 6 · 2.54 mm · straight	Through-hole	**C37208** 2.54-1x6P	Extended	1	1 +3V3 · 2 SWCLK · 3 GND · 4 SWDIO · 5 NRST · 6 SWO
Sheet_4 Signs of life	An LED, a button and a serial port, on the same pins as the Nucleo					
☐ **R4**	**LED resistor** 1 kΩ · 1%	0603	**C21190** 0603WAF1001T5E	Basic	1	LED (PA5, pin 21) to D1 anode
☐ **D1**	**User LED** Red	0603	**C2286** KT-0603R	Basic	1	D1 cathode to GND
☐ **SW2**	**User button** Tactile, 5.1 × 5.1 mm	SMD-4P	**C318884** TS-1187A-B-A-B	Check	1	BUTTON (PC13, pin 2) to GND. Pull-up on in CubeMX
☐ **J3**	**UART header** 1 × 3 · 2.54 mm · straight	Through-hole	**C49257** 2.54-1*3P	Extended	1	1 UART_TX (PA2) · 2 UART_RX (PA3) · 3 GND

Order list

20 different parts · 34 pieces per board · 10 Basic, 7 Extended, 3 to check

LCSC #	Part	Refs	JLC class	Qty
C69336	STM32F446RET6	U1	Extended	1
C14663	100 nF X7R	C1–C4, C5, C8, C10	Basic	7
C19666	4.7 µF X5R	C6, C9	Basic	2
C1002	Ferrite bead 600 Ω	FB1	Basic	1
C15849	1 µF X5R	C7	Basic	1
C25804	10 kΩ	R1	Basic	1
C318884	Tactile switch	SW1, SW2	Check	2
C165948	USB-C receptacle	J1	Extended	1
C23186	5.1 kΩ	R2, R3	Basic	2
C7519	USBLC6-2SC6 ESD	U3	Extended	1

LCSC #	Part	Refs	JLC class	Qty
C82942	ME6211 3.3 V LDO	U2	Extended	1
C19702	10 µF X5R	C15, C16	Basic	2
C21190	1 kΩ	R4, R5	Basic	2
C2286	Red LED	D1, D2	Basic	2
C115962	8 MHz crystal	Y1	Basic	1
C1656	27 pF C0G	C11, C12	Extended	2
C1985642	32.768 kHz crystal	Y2	Check	1
C59526	4.3 pF C0G	C13, C14	Check	2
C37208	1 × 6 header	J2	Extended	1
C49257	1 × 3 header	J3	Extended	1

Why 27 pF on the 8 MHz crystal

Each cap = $2 \times (C_L - C_{stray}) = 2 \times (20 - \text{about } 6.5) \approx 27$ pF.

The Basic 22 pF (**C1653**) also works. The clock runs a few tens of ppm fast, which USB and UART don't notice.

Will Y1 start? Yes

$g_{m,crit} = 4 \times ESR \times (2\pi F)^2 \times (C_0 + C_L)^2 \approx 0.43$–$0.59$ mA/V for C_0 of 3–7 pF.

The F446 allows up to 1 mA/V, so it passes. Y2 is the Nucleo's own LSE crystal with the same 4.3 pF caps; use LSE high drive.

If a part is out of stock

Search the same value and package, for example "100nF 0603 X7R", and pick a Basic part.

Keep the voltage rating the same or higher, and keep C0G on the crystal caps.

Your Nucleo-F446RE this week

Same chip as our board, so write and test the firmware now

Nucleo details from ST's user manual UM1724 (board MB1136).

Same pins on purpose

Code for the LED, button and serial runs on both boards

Function	Nucleo-F446RE	Our board
User LED	LD2 on PA5	D1 on PA5, net `LED`
User button	B1 on PC13, pressed = low	SW2 on PC13, net `BUTTON`. Turn on the pull-up in CubeMX
Serial port	USART2 on PA2, PA3, through the ST-LINK as a USB COM port	USART2 on PA2, PA3, out to J3. Use a 3.3 V USB-serial adapter
8 MHz clock	No crystal fitted (X3). The ST-LINK feeds 8 MHz into PH0. CubeMX: HSE **BYPASS Clock Source**	Y1 crystal. CubeMX: HSE **Crystal/Ceramic Resonator**
32.768 kHz	X2: ABS25-32.768KHZ-6-T with 4.3 pF caps	Y2: the same crystal and caps
Debug header	CN4: 1 VDD_TARGET · 2 SWCLK · 3 GND · 4 SWDIO · 5 NRST · 6 SWO	J2: the same order, so each wire goes pin 1 to pin 1, and so on

The one firmware change between the boards is the HSE setting in CubeMX's RCC page. Forget it and our board's crystal never starts.

Program our board with the Nucleo's ST-LINK

1. **Take off both CN2 jumpers.** That disconnects the ST-LINK from the Nucleo's own chip.
2. **Wire CN4 to J2,** pin 1 to pin 1 through pin 6 to pin 6.
3. **Power our board from its own USB-C.** CN4 pin 1 only senses our 3.3 V; it doesn't supply it.
4. **Connect in STM32CubeProgrammer or CubeIDE over SWD.** If it won't connect, choose "Connect under reset" (that's why NRST is on pin 5).
5. **Put both CN2 jumpers back** to program the Nucleo again.

First firmware, on the Nucleo

1. New STM32 project, board **NUCLEO-F446RE**.
2. Blink PA5 with **HAL_GPIO_TogglePin** and a 500 ms **HAL_Delay**.
3. Read PC13: the LED changes when B1 is pressed (it reads 0).
4. Send "hello" with **HAL_UART_Transmit** on USART2 at 115200 baud and read it in a serial terminal.
5. Save it. This becomes the bring-up test for our board.

Plan to Thursday, then the final

Tick each box as it's done

When	Schematic in EasyEDA	Nucleo
Sat–Sun Oct 31–Nov 1	☐ Project made with four sheets. ☐ Every part placed by its C-number. ☐ Sheet_1 wired and labelled.	☐ STM32CubeIDE installed. ☐ LD2 blinking.
Mon Nov 2	☐ Sheet_2: USB-C, ESD, regulator. ☐ Sheet_3: crystals and J2.	☐ B1 toggles LD2.
Tue Nov 3	☐ Sheet_4 wired. ☐ Every pin checked against the build sheet. ☐ Design Manager (`Ctrl` + `D`) shows no stray nets.	☐ "hello" over USART2 at 115200 in a serial terminal.
Wed Nov 4	☐ Swap schematics with a partner and check each other's. ☐ Fixes made. ☐ Convert to PCB and start placing.	☐ Same firmware set to HSE Crystal, ready for our board.
Thu Nov 5	**Checkpoint:** all four sheets printed and checked.	**Demo:** LED, button and serial on the Nucleo.
Week 2 Nov 9–13	☐ Layout, routing, ground pour. ☐ DRC clean. ☐ 3D view checked. ☐ Order early in the week if you want boards for the final: making and shipping can take one to two weeks.	
Week 3 Final	**Bring-up** if boards are back: 3.3 V on every VDD pin first, then SWD connects, then blink.	**Or present** the schematic, layout and 3D view with the Nucleo demo.

STM32F446 board: from the basics up		Rev A
Title: **Nucleo and the plan**	Editor: **EasyEDA Std**	Sheet: **3 of 3**